



Modelling Financial Markets

Seminar 15: LLMs and Sentiment Analysis — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- AI: Artificial Intelligence
- AUC: Area Under the (ROC) Curve
- BERT: Bidirectional Encoder Representations from Transformers
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAR: Cumulative Abnormal Return
- CI: Confidence Interval
- ECB: European Central Bank
- FNSPID: Financial News and Stock Price Integration Dataset
- GI: (Harvard) General Inquirer, dictionary IV-4
- LLM: Large Language Model
- LM: Loughran-McDonald (financial sentiment dictionary)
- OLS: Ordinary Least Squares
- ROC: Receiver Operating Characteristic (curve)
- SE: Standard Error
- TF-IDF: Term Frequency - Inverse Document Frequency
- US: United States
- UTC: Coordinated Universal Time (Temps Universel Coordonné)

Outline

Introduction

Part A: Pen and Paper

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Score a text with a sentiment dictionary and with the label probabilities of a large language model (LLM)
- ▣ Read a confusion matrix: accuracy, precision, recall, F1 and macro-F1
- ▣ Compare two classifiers with a paired test and detect training-data leakage and look-ahead bias
- ▣ Turn headline sentiment into a trading signal with correct timing, clustered standard errors and costs

- ▣ Background: Lecture 15; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight pen-and-paper exercises (A1–A8)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** one open question, a seed for the team project (C1), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- You may use an AI assistant, but declare it in [AI_USE.md](#) and log the errors you catch in [AI_ERRORS.md](#)
 - ▶ C2 trains exactly this: find the errors in an AI answer about sentiment backtests
- Always ask: was the text public before the return I try to predict, and did the model already know the outcome?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (tone with two dictionaries), A3 (metrics from a confusion matrix), A5 (label probabilities and temperature), A7 (break-even cost)
- Part B: B1 (FinBERT vs LM, McNemar), B3 (training-data leakage), B5 (clustered standard errors), B7 (does the LLM remember the market?)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A6 (majority class; model: A3), A8 (longer holding; model: A7)
- Part B: B2 (LLM vs FinBERT; model: B1), B4 (how many labels; model: B3), B6 (event study; model: B5), B8 (prompt sensitivity; model: B1)
- Part C: C1, does headline sentiment pay after costs?; C2, critique an AI answer on an LLM sentiment backtest

Setup: Python Environment and Data

-  **Open in Colab:** [chapter15_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `scikit-learn`, `matplotlib`; `transformers` only to recompute the scores
- Labelled texts: Financial PhraseBank (4 846 news sentences, 2 264 with full agreement) [Malo et al. \(2014\)](#); Twitter Financial News, validation set (2 388 headlines)
- Dated headlines: FNSPID (Financial News and Stock Price Integration Dataset) [Dong, Fan & Peng \(2024\)](#), 106 820 headlines on 16 large US stocks; daily closes of the stocks and of SPY (an S&P 500 index fund)
- The scores of every model are precomputed in the tables of Chapter 15; Quantlets: [MFM_ch15_sem_partA](#), [partB](#), [partC](#)

Formulas Used Today (1/2): Scores and Metrics

- Dictionary tone: $\tau = (P - N)/(P + N)$, P , N : counts of positive and negative words; $\tau = 0$ if $P + N = 0$
 - ▶ LM: the Loughran–McDonald financial word lists [Loughran & McDonald \(2011\)](#); GI: the Harvard General Inquirer IV-4 list, a generic psychology dictionary
- LLM label probabilities: $p_k = e^{z_k/T} / \sum_j e^{z_j/T}$ (z_k : logit of answer word k , T : temperature); net score $s = p_{\text{pos}} - p_{\text{neg}}$
 - ▶ FinBERT: a BERT (Bidirectional Encoder Representations from Transformers) model fine-tuned for financial sentiment [Araci \(2019\)](#); Qwen2.5: open-weight LLMs [Qwen Team \(2024\)](#)
- Class k : precision = $\frac{TP_k}{TP_k + FP_k}$, recall = $\frac{TP_k}{TP_k + FN_k}$, $F1_k = \frac{2 \text{ prec} \cdot \text{rec}}{\text{prec} + \text{rec}}$; macro-F1 = $\frac{1}{3} \sum_k F1_k$
 - ▶ TP_k : texts of class k labelled k ; FP_k : other texts labelled k ; FN_k : texts of class k labelled otherwise

Formulas Used Today (2/2): Inference and Signals

- McNemar test [McNemar \(1947\)](#): n_{01} texts where A is wrong and B right, n_{10} the reverse; under H_0 , $n_{01} \sim \text{Bin}(n_{01} + n_{10}, 0.5)$
 - Bootstrap CI (confidence interval): resample texts 2 000 times, take the 2.5% and 97.5% quantiles
- AUC (area under the ROC curve): probability that a random “up” case gets a higher score than a random “down” case; 0.5 = no information
- Panel regression $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon$: r^e excess return over SPY (%), z standardised daily score, $h = 0, 1, 2$ days after the news day d
 - Standard errors (SE): ordinary least squares (OLS) or clustered by day or by stock [Petersen \(2009\)](#); bp: basis point, 0.01%
- Strategy: Newey–West t [Newey & West \(1987\)](#); annualised Sharpe ratio $\sqrt{252} \bar{r}/s_r$; break-even cost = mean gross return on days with positions / 4 (four trades a day)

A1: One Sentence, Two Dictionaries [Solved]

Task

- Financial PhraseBank sentence, labelled **positive** by all annotators:
 - ▶ “Uponor improved its performance in spite of the decrease in residential construction in the US.”
- (a) Find the LM and the GI positive and negative words (15 tokens)
- (b) Compute the tone τ and the class for each dictionary
- (c) Why does the generic dictionary fail here?

Solution

- LM: positive *improved*; negative –
 - ▶ $\tau_{LM} = (1 - 0)/(1 + 0) = +1.00$: positive, correct
- GI: positive –; negative *spite*, *decrease*
 - ▶ $\tau_{GI} = (0 - 2)/(0 + 2) = -1.00$: negative, wrong
- GI misses *improved* and counts *spite* and *decrease*: they describe the housing market, not the firm
- A bag of words cannot see that “in spite of” reverses the role of what follows

A2: A Neutral Sentence [Proposed]

Solution discussed in class

Task

- Financial PhraseBank sentence, labelled **neutral** by all annotators:
 - ▶ “The ECB can mainly target inflation.”
- (a) LM and GI words, tone and class
- (b) Which dictionary agrees with the annotators, and why?
- (c) Is *inflation* bad news for a bank? For a bond holder?
- Model: A1

A3: Metrics from a Confusion Matrix [Solved]

Task

- FinBERT on the Twitter validation set:

True \ predicted	neg	neu	pos	Total
neg	264	71	12	347
neu	213	1 184	169	1 566
pos	58	133	284	475
Total	535	1 388	465	2 388

- (a) Accuracy
- (b) Precision, recall and F1 for each class
- (c) Macro-F1; which class is the weakest, and why?

Solution

- Accuracy = $(264 + 1\,184 + 284) / 2\,388 = 72.5\%$
- Negative: prec $264 / 535 = 49.3\%$, rec $264 / 347 = 76.1\%$, F1 59.9%
- Neutral: prec 85.3% , rec 75.6% , F1 80.2%
- Positive: prec $284 / 465 = 61.1\%$, rec 59.8% , F1 60.4%
- Macro-F1 = $(59.9 + 80.2 + 60.4) / 3 = 66.8\%$, below the accuracy
- Negative is weakest in precision: 213 neutral headlines are called negative; a short strategy would trade on them

A4: The Loughran–McDonald Classifier [Proposed]

Solution discussed in class

Task

- The LM dictionary on the same headlines:

True \ predicted	neg	neu	pos	Total
neg	172	166	9	347
neu	247	1 192	127	1 566
pos	45	349	81	475
Total	464	1 707	217	2 388

- (a) Accuracy, precision, recall and F1 per class, macro-F1
- (b) Which class does LM miss most, and why?
- Model: A3

A5: From Label Probabilities to a Score [Solved]

Task

- Qwen2.5-7B, prompt P1, one Twitter headline labelled **neutral** by the annotators
 - ▶ $p_{\text{neg}} = 0.608$, $p_{\text{neu}} = 0.392$, $p_{\text{pos}} = 0.000$ (rounded)
- (a) Predicted label and net score s ; is the label right?
- (b) Logit gap $z_{\text{neu}} - z_{\text{neg}}$
- (c) Probabilities, label and score at temperature $T = 2$

Solution

- Label $\arg \max_k p_k = \text{negative}$: wrong;
 $s = 0.000 - 0.608 = -0.608$
- $z_{\text{neu}} - z_{\text{neg}} = \ln(p_{\text{neu}}/p_{\text{neg}}) = \ln(0.392/0.608) = -0.437$
- At T : $p_k(T) \propto p_k^{1/T}$, here $\sqrt{p_k}$: 0.780, 0.626, 0; sum 1.406
 - ▶ $p(T) = (0.554, 0.446, 0.000)$, $s(T) = -0.554$
- Temperature is monotone: the label does not change, but the score shrinks towards 0; thresholds on s then select different headlines

A6: The Majority-Class Benchmark [Proposed]

Solution discussed in class

Task

- Twitter validation set: 1 566 neutral, 475 positive, 347 negative headlines ($N = 2\,388$)
- A “classifier” always answers *neutral*
- (a) Its accuracy
- (b) Precision, recall and F1 of each class; macro-F1
- (c) Which yardstick would you report for the LM classifier of A4, and why?
- Model: A3

A7: The Break-Even Cost of a Daily Signal [Solved]

Task

- FinBERT strategy of Lecture 15: at the close of $d + 1$, buy stocks with positive and sell those with negative news score of day d , each hedged with SPY; close one day later
- Days with positions: 96.2%; mean gross excess return on those days: 3.40 bp
- (a) Trades per day with positions; break-even cost per trade
- (b) Mean net return per day with positions at 5 bp per trade
- (c) How much larger would the gross edge have to be?

Solution

- Four trades: stock and SPY, in and out; break-even $c^* = 3.40/4 = 0.85$ bp
- Net: $3.40 - 4 \cdot 5 = -16.60$ bp per day with positions
- At 5 bp the gross edge must reach 20 bp a day: $20/3.40 = 5.9$ times more
- Daily turnover is the enemy: costs are paid every day, the edge is a few basis points

A8: Holding for Five Days [Proposed]

Solution discussed in class

Task

- ▣ Same edge as A7, 3.40 bp per day, but each position is held 5 days: one round trip (four trades) every 5 days
- ▣ (a) Cost per day at 5 bp per trade; break-even cost per trade
- ▣ (b) Is it realistic to assume the same daily edge on days $d + 2$ to $d + 6$?
- ▣ Model: A7

B1: Is FinBERT Better than LM? [Solved]

Task

- Twitter validation set, 2 388 headlines; labels of the LM dictionary and of FinBERT
- (a) Accuracy of each with a bootstrap 95% CI
- (b) Difference FinBERT – LM with a paired bootstrap CI (the same resampled texts for both)
- (c) Exact McNemar test [McNemar \(1947\)](#)
- Code: `b_pairs` in the notebook

Interpretation

- Why is the paired test more powerful than comparing the two intervals?
- Which texts carry no information in McNemar's test?
- Is a difference of this size useful for trading?

B1: Results [Solved]

	Accuracy (%)	CI 95%
LM	60.5	[58.5; 62.4]
FinBERT	72.5	[70.7; 74.4]
Difference (pp)	+12.0	[+9.8; +14.2]

- n_{01} (LM wrong, FinBERT right) = 539, $n_{10} = 252$; under H_0 we expect 395.5 each; exact $p < 0.001$
- The large-sample version $(n_{01} - n_{10})^2 / (n_{01} + n_{10}) = 104.1$ vs χ_1^2 gives the same verdict
- Texts on which both are right or both wrong cancel out; pairing removes the variation of text difficulty
- A clear win, outside FinBERT's training data; pp: percentage points

B2: Is a 7-Billion-Parameter LLM Better than FinBERT? [Proposed]

Solution discussed in class

Task

- ▣ Same data; FinBERT against Qwen2.5-7B, zero-shot, prompt P1 (no Twitter labels used by either)
- ▣ (a) Accuracies with CI, the paired difference with CI
- ▣ (b) Exact McNemar test
- ▣ Model: B1

Interpretation

- ▣ Do the two intervals overlap? Does that settle the question?
- ▣ Qwen2.5-7B has far more parameters and is much slower: what would you still gain from it?

B3: Training-Data Leakage [Solved]

Task

- FinBERT was fine-tuned on Financial PhraseBank [Araci \(2019\)](#)
- (a) Accuracy and macro-F1 of FinBERT, Qwen2.5-7B, LM and MiniLM + logistic regression on
 - ▶ PhraseBank, full agreement (2 264); all of PhraseBank (4 846); Twitter (2 388)
- (b) Drop from PhraseBank to Twitter for each model
- MiniLM [Wang et al. \(2020\)](#): sentence embeddings; its logistic regression is trained on the Twitter training set

Interpretation

- Twitter headlines are harder for everyone: how do you separate “harder data” from “memory of the training set”?
- Why does the MiniLM model move the other way?

B3: Results [Solved]

Accuracy (%)	PhraseBank, 100% agr.	PhraseBank, all	Twitter	Drop (pp)	Macro-F1, Twitter
FinBERT	97.2	89.0	72.5	16.4	66.8
Qwen2.5-7B	93.6	81.4	74.2	7.2	72.6
LM	65.0	62.5	60.5	2.0	46.2
MiniLM + log. reg.	81.4	70.6	78.6	−8.0	69.9

- Drop: all of PhraseBank minus Twitter; FinBERT CI on PhraseBank [88.1; 89.8], on Twitter [70.7; 74.4]
- Difference in differences: FinBERT loses 9.2 pp more than Qwen2.5-7B, which never saw PhraseBank labels: a measure of the leakage
- MiniLM is trained on Twitter: better at home, worse on PhraseBank; every model is best on its own training domain

 MFM_ch15_sem_partB

B4: How Many Labels Do You Need? [Proposed]

Solution discussed in class

Task

- ▣ Logistic regression on MiniLM embeddings and on TF-IDF features (term frequency–inverse document frequency: word and word-pair counts, down-weighted if frequent in all texts)
- ▣ Trained on random subsets of the Twitter training set ($n = 100$ to $9\,543$), 10 repetitions each; evaluated on the validation set
- ▣ (a) Smallest n at which each beats FinBERT (72.5%)
- ▣ Model: B3

Interpretation

- ▣ Why do embeddings start higher but grow more slowly?
- ▣ How many texts would you label for a new market, e.g. Romanian stock news?

B5: Reaction or Prediction, and Which Standard Errors? [Solved]

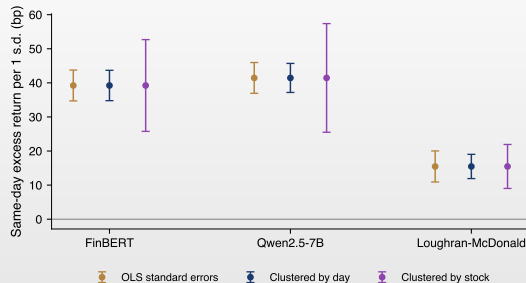
Task

- Panel of 18 282 stock-days with news (3 434 trading days, 16 stocks); daily score = mean of the headline scores
- (a) Regress r_{d+h}^e on z_d for FinBERT, Qwen2.5-7B and LM, $h = 0, 1, 2$
- (b) Standard errors: OLS, clustered by day, clustered by stock [Petersen \(2009\)](#)
- FNSPID gives only the date: day d is the first trading day on or after it, and $d + 2$ is the first return surely after publication

Interpretation

- Why do the conclusions for $h = 0$ and $h = 2$ differ?
- Which clustering matters here, and can you trust a clustered SE with 16 clusters?

B5: Results (1/2) [Solved]



- Slope b (bp of excess return per 1 standard deviation of score) at $h = 0$, with 95% intervals from the three standard errors

B5: Results (2/2) [Solved]

Score	h	b (bp)	SE OLS	SE day	SE stock	t day	t stock
FinBERT	0	+39.2	2.3	2.3	6.9	+17.27	+5.71
FinBERT	2	-0.5	2.2	2.2	1.5	-0.25	-0.35
Qwen2.5-7B	0	+41.4	2.3	2.2	8.1	+19.07	+5.10
Qwen2.5-7B	2	+2.3	2.2	1.9	1.0	+1.19	+2.41
LM	0	+15.5	2.3	1.8	3.3	+8.53	+4.70
LM	2	-0.0	2.2	1.7	0.9	-0.01	-0.03

- $h = 0$: large and significant under every SE: headlines describe the move of the day, often the move itself ("shares jump")
- $h = 2$: FinBERT -0.5 bp ($t = -0.25$): the tradable forecast is not there
- By day $\approx$ OLS (only 5.3 stocks per day); by stock the SE of FinBERT at $h = 0$ is 3.0 times larger: the slope differs across stocks
 - With 16 clusters the clustered SE is itself noisy: use t_{15} critical values or a wild cluster bootstrap [Cameron, Gelbach & Miller \(2008\)](#); Qwen at $h = 2$ ($t = +2.41$ by stock) is one of 9 tries

B6: Event Study with the Loughran–McDonald Tone [Proposed]

Solution discussed in class

Task

- Split news days into terciles of the daily LM tone; repeat with FinBERT
- For the most negative and the most positive third, average across stock-days of each date the excess returns on days -5 to $+5$
- (a) CAR (cumulative abnormal return) before the news (-5 to -1), day 0, day 1, after ($+2$ to $+5$), with Newey–West t
- Model: B5

Interpretation

- Does the LM tone separate winners from losers on day 0 as well as FinBERT does?
- Why is the post-event return positive for both groups? Tetlock (2007)

B7: Does the Model Remember the Market? [Solved]

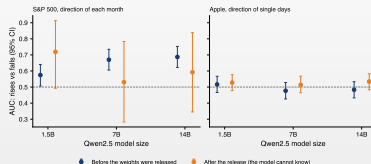
Task

- Prompt without context: “Did the S&P 500 index rise or fall during ⟨month year⟩?”; score $P(\text{rise})$ from the answer-word probabilities
- Qwen2.5 1.5B, 7B, 14B; weights published on 19 September 2024 [Qwen Team \(2024\)](#)
- (a) AUC for the direction of each month, before (to August 2024) and after (from October 2024) the release, with bootstrap CI
- (b) Permutation p -value for 14B before the release (2000 shuffles of the up/down labels)

Interpretation

- What does an AUC above 0.5 before the release mean, if the model sees no data?
- Can the post-release sample prove that the memory is gone?
- What does this imply for backtests that score old news with an LLM? [Glasserman & Lin \(2023\)](#)

B7: Results [Solved]



Qwen2.5	AUC before	CI 95%	AUC after	CI 95%	$\bar{P}(\text{rise})$ up / down
1.5B	0.57	[0.51; 0.64]	0.72	[0.49; 0.91]	0.36 / 0.30
7B	0.67	[0.61; 0.73]	0.53	[0.28; 0.78]	0.73 / 0.43
14B	0.69	[0.62; 0.75]	0.59	[0.35; 0.84]	0.81 / 0.54

- Before: 296 months (185 up, 111 down); after: 23 months; permutation p (14B) < 0.001
- 7B and 14B “know” the monthly direction of the past: no data were given, so the signal is memory of the training text
- After the release the intervals include 0.5, but 23 months are too few to prove the memory is gone

B8: Prompt Sensitivity [Proposed]

Solution discussed in class

Task

- Three prompts that ask the same question, Qwen2.5-7B and 14B, Twitter validation set
 - ▶ P1: “classify the sentiment . . . for investors”; P2: “good, bad or neutral for the stock price”; P3: “what is the sentiment of this text”
- (a) Accuracy with CI and share of *neutral* answers for each prompt
- (b) McNemar P1 vs P3; share of headlines with the same label under all three prompts
- Model: B1

Interpretation

- If you picked the best prompt on this set, what would you be doing?
- Why does the best prompt also answer *neutral* most often?

C1: Does Headline Sentiment Pay after Costs?

- Strategy of Lecture 15: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, equal weights; close at $d + 2$
 - ▶ Days without news: no position, return 0; costs: 5 bp per trade, four trades a day
- (a) Mean daily gross return, Newey–West t , block-bootstrap 95% CI (blocks of 10 days)
- (b) Sharpe ratio before and after costs; break-even cost; Sharpe for costs of 0 to 10 bp
- (c) Strong news only: trade only if $|s_{i,d}| > 0.5$; (d) the same for Qwen2.5-7B and LM
- Discuss: timing, look-ahead of the LLM (B7), survivorship of the 16 stocks, crowding [Lopez-Lira & Tang \(2026\)](#)
- Deliverable: a two-page note with charts, and a Quantlet folder (code, Metainfo, charts)

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

- **Student prompt:** “I have FNSPID headlines for 16 US stocks, 2009–2023, and daily excess returns vs SPY. Check that Qwen2.5-7B reads sentiment, then test whether its headline sentiment forecasts returns out of sample.”
- **Answer of the AI assistant** (qwen_label, qwen_score: the zero-shot scorer of Lecture 15)

```
import pandas as pd, statsmodels.formula.api as smf
from sklearn.metrics import accuracy_score
tw['pred'] = [qwen_label(t) for t in tw['text']] # Twitter validation set
acc = accuracy_score(tw['label'], tw['pred'])
news['day'] = first_trading_day_on_or_after(news['date'])
news['s'] = [qwen_score(t) for t in news['title']] # P(pos) - P(neg)
d = news.groupby(['ticker', 'day'])['s'].mean().reset_index()
d['z'] = (d['s'] - d['s'].mean()) / d['s'].std()
d = d.merge(rets, on=['ticker', 'day']) # rets: ticker, day, xret (%)
fit = smf.ols('xret ~ z', data=d).fit(cov_type='cluster',
                                     cov_kws={'groups': pd.factorize(d['day'])[0]})
```

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

□ Conclusion of the AI assistant

- ▶ "Accuracy 74.2%, more than twice the 33% of a random guess among three classes: Qwen2.5-7B reads financial sentiment very well."
- ▶ "Slope +41.4 bp per standard deviation of the score, $t = +19.07$ (clustered by day): the day's headlines forecast the stock's excess return."
- ▶ "Qwen2.5 was never trained on these labels or return series, so the 2009–2023 backtest is fully out of sample."

□ The answer contains **three** errors typical for sentiment backtests; the code runs and the numbers are right

- ▶ Find each error: which line or sentence, and why it is wrong
- ▶ Say how you would detect it without knowing the answer: a class count, a timestamp check, a release date, a memory test
- ▶ Give the corrected code or statement
- ▶ Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

Common Pitfalls

- ▣ Reporting accuracy without the majority-class benchmark and macro-F1
- ▣ Evaluating a model on its own training data (FinBERT on PhraseBank)
- ▣ Comparing two classifiers through separate intervals instead of a paired test
- ▣ Matching a date-only headline to the same-day return and calling it a forecast
- ▣ Backtesting an LLM on news older than the release of its weights
- ▣ Choosing the prompt, the threshold or the model after seeing the test results
- ▣ Ignoring costs and the number of clusters: four trades a day, only 16 stocks

Take-Home Assignment (Team Project)

- Choose a text source not covered in class: BVB (Bucharest Stock Exchange) announcements, Romanian financial news, central-bank statements or crypto posts
- Label at least 200 texts yourselves (two annotators, report the agreement)
- Compare LM, FinBERT, an open LLM and embeddings + logistic regression: accuracy and macro-F1 with CI, McNemar tests
- Build a signal with correct timing; report the period after the LLM's release separately
- Declare AI use in [AI_USE.md](#) and log every AI error you catch in [AI_ERRORS.md](#) (at least three per project, as in C2)
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Natural Language Processing
- ▣  The Next Word Problem
- ▣  DELTA
- ▣  LDA Extensions

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_15

References (1/2)

- ▣ Araci, D. (2019). [FinBERT: financial sentiment analysis with pre-trained language models](#). [arXiv:1908.10063](#).
- ▣ Cameron, A.C., Gelbach, J.B., Miller, D.L. (2008). [Bootstrap-based improvements for inference with clustered errors](#). *Review of Economics and Statistics*, 90(3), 414–427.
- ▣ Dong, Z., Fan, X., Peng, Z. (2024). [FNSPID: a comprehensive financial news dataset in time series](#). *Proceedings of the 30th ACM SIGKDD Conference*, 4918–4927.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Glasserman, P., Lin, C. (2023). [Assessing look-ahead bias in stock return predictions generated by GPT sentiment analysis](#). [arXiv:2309.17322](#).
- ▣ Lopez-Lira, A., Tang, Y. (2026). [Can ChatGPT forecast stock price movements? Return predictability and large language models](#). *Journal of Financial Economics*, 184, 104335.
- ▣ Loughran, T., McDonald, B. (2011). [When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks](#). *Journal of Finance*, 66(1), 35–65.
- ▣ Malo, P., Sinha, A., Korhonen, P., Wallenius, J., Takala, P. (2014). [Good debt or bad debt: detecting semantic orientations in economic texts](#). *Journal of the Association for Information Science and Technology*, 65(4), 782–796.

References (2/2)

- ▣ McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2), 153–157.
- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Petersen, M.A. (2009). Estimating standard errors in finance panel data sets: comparing approaches. *Review of Financial Studies*, 22(1), 435–480.
- ▣ Qwen Team (2024). Qwen2.5 technical report. arXiv:2412.15115.
- ▣ Sarkar, S.K., Vafa, K. (2024). Lookahead bias in pretrained language models. SSRN Working Paper 4754678.
- ▣ Tetlock, P.C. (2007). Giving content to investor sentiment: the role of media in the stock market. *Journal of Finance*, 62(3), 1139–1168.
- ▣ Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., Zhou, M. (2020). MiniLM: deep self-attention distillation for task-agnostic compression of pre-trained Transformers. *NeurIPS*; arXiv:2002.10957.